Journal of Economic Behavior and Organization 152 (2018) 1–16

Contents lists available at ScienceDirect

Journal of Economic Behavior and Organization

journal homepage: www.elsevier.com/locate/jebo

An empirical analysis of taxi, Lyft and Uber rides: Evidence from weather shocks in NYC☆

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ARTICLE INFO

Article history:
Received 21 August 2017
Revised 4 June 2018
Accepted 5 June 2018

JEL classification:
D01
D03
L92
J22

Keywords:
Rain
Taxi
Uber
Lyft
Dynamic pricing

ABSTRACT

Using all taxi, Lyft and Uber rides in New York City, we show that the number of Uber and Lyft rides is significantly correlated with whether it rained. The number of Uber (Lyft) rides per hour is about 22 (19)% higher when it is raining, while the number of taxi rides per hour increases by only 5% in rainy hours-suggesting that surge pricing (prime time) encourages an increase in supply. We show that while the number of taxi rides, passengers and fare income all significantly decreased after Uber entered the market in May 2011, taxis do not respond differently to increased demand in rainy hours than non-rainy hours since the entrance of Uber. Last, we test whether Lyft's entry in the market affected Uber. Our estimates suggest that Uber was still growing after Lyft entered the market, but that Uber rides during rainy hours decreased by about 9%. Our findings suggest that dynamic pricing make Lyft and Uber drivers compete for rides when demand suddenly increases, i.e., during rainy hours.

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1. Introduction

Standing or walking in the rain is an activity best avoided. In New York City (NYC), when faced with such inclement weather, the demand for personal transportation naturally increases. During such scenarios, taxi drivers spend less time searching for customers and could thus earn a higher wage. Nonetheless, it has been a common complaint that it is difficult to find a taxi in the rain.

☆ An earlier version of this paper circulated under the title: "Has Uber Made it Easier to Get a Ride in the Rain." We are grateful to the editor, two referees, and Cristina Blanco-Perez, Pierre Brochu, Henry S. Farber, Jason Garred, Anthony Heyes, Kim P. Huynh, Louis-Philippe Morin, Matt Webb and seminar participants at the 2018 ASSA annual meeting, Bank of Canada and SCSE for very useful remarks and encouragement. We thank Mitch Prior for outstanding research assistance. The paper was written while Kerry Nield was completing graduate studies at Carleton University. The views expressed in this paper are those of the authors. No responsibility for them should be attributed to the Bank of Canada. All errors are our own.

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https://doi.org/10.1016/j.jebo.2018.06.004
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Previous explanations have considered that the dearth in availability may be due to daily income targeting behavior[1] by taxi drivers: on rainy days taxi drivers reach their target faster and work less hours.[2] This view has been rebutted by Farber (2015). In this recent study, he rejects the idea that target earnings behavior explains the difficulty to find taxis when it rains. Instead, Farber (2015) provides evidence that some drivers stop working because of the worsening driving conditions associated with rain; unfavorable working conditions for which there is no compensating differential. Taxi drivers face many market frictions: the rates for rides are fixed and the supply of vehicles is controlled.

As alternative services to taxis, Uber and Lyft entered the NYC market respectively in May 2011 and July 2014 with dynamic pricing and mobile driver-passenger matching technology. Surge pricing and prime time are pricing schemes respectively implemented by Uber and Lyft where passengers pay a higher rate for the ride during times of high demand; this higher pricing scheme gives incentives to drivers to provide rides in inclement conditions. Furthermore, unlike taxis which have a regulated supply by the medallion system, Uber and Lyft's mobile app based service allows almost anyone with access to a vehicle to operate as an Uber or Lyft driver. Uber and Lyft could thus be logical responses to unmet demand during adverse conditions.

Previous studies on intertemporal substitution of labor supply find mixed evidence of intertemporal substitution effects when individuals face increased earning opportunities. (See Blundell and MaCurdy (1999) for a review of the literature.) Fehr and Goette (2007) conduct a field experiment at a bicycle messenger service in Switzerland in which they vary temporarily the piece rate paid to messengers. They find a large positive elasticity of hours of work, but a negative elasticity of effort per hour. In another experiment, Chang and Gross (2014) document that pear packers in California respond to unexpected overtime by decreasing their effort. In Oettinger (1999), the author studies labor supply of stadium vendors at baseball games and finds that the number of vendors who worked in a game is positively related to changes in expected wages.

In contrast to earlier studies (e.g., Chen and Sheldon (2015)), we have the unique opportunity of comparing the response to exogenous circumstances of three groups of workers, taxis, Lyft and Uber in NYC, who are subject to different market structures. Another key feature of our analysis is that we can test whether the market changed since Uber and Lyft entered the on-demand transportation network in NYC.

This paper makes at least four contributions. First, we provide an empirical examination of whether the number of Uber and Lyft rides increases more than the number of taxi rides when it rains. For this analysis, we rely on all taxi, Lyft and Uber rides in NYC from April–September 2014 and January 2015 to December 2016 and merge this data set with weather data from the National Weather Service Observatory in Central Park. The results suggest that the number of Uber and Lyft rides per hour is significantly correlated with whether it rained. More precisely, the number of Uber (Lyft) rides per hour is about 22 (19)% higher when it is raining, suggesting that dynamic pricing encourages an increase in supply.[3] This increase in transactions when it rains likely reflects a large increase in both demand and supply. On the other hand, the number of taxi rides per hour increases by only 5% in rainy hours during this time period. This is indicative that the market frictions faced by taxis are discouraging a larger response to increased demand. These results suggest that Uber and Lyft drivers are more responsive to rain than taxis and provide suggestive evidence that it is partly because of the increase in fare coming from dynamic pricing.

Second, we test whether rain increases the number of *daily* rides. We provide evidence that the number of Uber rides increases by approximately 10% during rainy days and that an additional hour of rain increases the number of Uber rides per day by approximately 2%. On the other hand, there is weak evidence that the daily number of taxi rides is related to rain. The number of taxi rides increases by approximately 2% during rainy days. This is suggestive evidence that Uber drivers, like taxi drivers, do not have a daily target income level and that Uber driver labor supply is best characterized by the neoclassical model.

Third, we use data for all trips taken in NYC taxi cabs before (January 2010–April 2011) and after (2014–2016) Uber and Lyft's rise in popularity, and show that the number of taxi rides per hour decreased by approximately 25% after Uber entered the New York market in May 2011. The results indicate that hourly fare income and the number of passengers per hour for taxis significantly decreased after Uber entered the market. Moreover, we show that the number of taxi rides in both rainy and non-rainy hours have similarly decreased since Uber entered the market, suggesting that taxis are not responding differently to increased earning opportunities since the entrance of new competitors.

Finally, we test whether Lyft's entry in the market affected Uber. Our estimates suggest that Uber was still growing after Lyft entered the market, but that Uber rides during rainy hours decreased by about 9%. Our findings suggest that dynamic pricing makes Lyft and Uber drivers compete for rides when demand suddenly increases, i.e., during rainy hours.

[1] See Koszegi and Rabin (2006) for a model of reference-dependent preferences and loss aversion. They build on the work of Kahneman and Tversky (1979) and Tversky and Kahneman (1991) and develop a model where a person's utility depends on a reference point.

[2] In a seminal article, Camerer et al. (1997) present evidence suggesting that taxi drivers have income reference-dependent preferences, so that taxi drivers have a daily income target and stop working once they reached this target. Note that Camerer et al. (1997) were not particularly interested in rain and did not claim that on rainy days taxi drivers reach their target faster and work less hours. We thank Colin Camerer for pointing this out.

[3] Unfortunately, our data set does not allow us to test whether the increase in labor supply of Uber and Lyft drivers during rainy hours is due to the extensive or intensive margin.

This paper contributes to the literature analyzing labor supply of taxi drivers (Agarwal et al., 2015; Buchholz, 2016; Camerer et al., 1997; Chou, 2002; Crawford and Meng, 2011; Doran, 2014; Farber, 2008; Haggag et al., 2017; Haggag and Paci, 2014; Martin, 2017). Two relevant papers using NYC data are Farber (2005) and Ashenfelter et al. (2010). Farber (2005) studies the stopping behavior of NYC taxi drivers and finds that the decision to stop working is primarily related to cumulative daily hours. Ashenfelter et al. (2010) also study the labor supply of taxi drivers. They analyze the impact of two permanent fare increases and find an elasticity of labor supply of −0.2 in response to these fare increases.

We also contribute to a small literature on rideshare drivers. To our knowledge, no empirical studies have studied Lyft and analyzed its impact on taxis and Uber. The dearth of research on this issue is mainly driven by the lack of publicly available data. Three recent papers using different Uber data are Chen and Sheldon (2015), Cohen et al. (2016) and Cramer and Krueger (2016). In Chen and Sheldon (2015), the authors use a randomly drawn subset of UberX driver-partners in a few American cities.[4] They find that UberX partners are less likely to quit at times with high surge prices. Cramer and Krueger (2016) compare capacity utilization for taxis and UberX drivers in five American cities. They provide evidence that the capacity utilization rates of taxi and UberX drivers are very similar in NYC.[5] They point out that this may be due to the high population density which supports more efficient matching of taxis. Last, Cohen et al. (2016) estimate consumer surplus for UberX in four U.S. cities. They rely on Uber's surge pricing algorithm to estimate demand elasticities and find that demand is quite inelastic (i.e., price elasticities between −0.4 and −0.6).

The paper is structured as follows. In the following section, we present competing theories of labor supply and briefly discuss the Lyft and Uber platforms. In Section 3, we detail the data sets and provide descriptive statistics. Section 4 presents the model specification. Section 5 discusses our findings of the impact of rain on taxi, Lyft and Uber rides. The last section concludes.

2. Conceptual framework

This section provides a framework for understanding how rain might impact the number of taxi, Lyft and Uber rides. We consider the standard neoclassical intertemporal models of labor supply (e.g., Fehr and Goette (2007) and Oettinger (1999)) and treat dynamic pricing as a transitory wage change for Uber and Lyft drivers. This is a contrast to earlier taxi literature that considers a behavioral model of reference-dependent preferences in which drivers stop working once they reach their target income level (e.g., Camerer et al. (1997) and Koszegi and Rabin (2006)).

2.1. Intertemporal models of labor

A simple neoclassical intertemporal model of labor predicts that workers respond positively to earnings opportunities. This model implies that unanticipated increases in earnings opportunities such as rain would positively affect labor supply elasticities. In other words, this model predicts that drivers would be working more on high-wage days and less on low-wage days.

A rational choice model with reference-dependent preferences suggests that workers who temporarily earn higher wages will eventually decrease their effort. For instance, cab and Uber drivers who have a reference level of daily income would work more in rainy hours. As they get closer to their daily reference level, the marginal utility of income would be decreasing which in turn would reduce their effort. The neoclassical model and the model with reference-dependent preferences that exhibit loss aversion around a daily target income level both suggest that rain increases hourly rides. On the other hand, only the former predicts that rain increases daily rides.

Using all trips taken in NYC taxis from 2009 to 2013, Farber (2015) tests whether rain affects earnings, utilization rates and the number of hacks on the streets. He finds that rain has no effect on hourly earnings, but does cause higher utilization of cabs, and in general shorter trips. This is indicative of increased demand for rides when it rains. However, there is also a decreased number of taxis on the streets which is, arguably, due to the disutility experienced by taxi drivers driving in uncomfortable weather conditions. Farber (2015) concludes that these results are "sufficient to reject the hypothesis that target earnings behavior contributes to the difficulty of finding a taxi in the rain [...]"

2.2. Dynamic pricing

Unlike taxis, the Uber and Lyft platforms adjust its prices using a realtime dynamic algorithm (Chen and Sheldon, 2015). Lyft and Uber fare rates automatically increase when demand is higher than supply of drivers within a fixed geographic area. Customers are informed of the higher fare before requesting the Lyft or Uber car. Uber (Lyft) drivers are also aware of surge pricing (prime time) and have access to maps displaying the locations currently surge pricing (prime time). Note that surge multipliers are discrete and there is no cap in NYC.

[4] UberX cars can carry up to four riders and have four doors. The full list of Uber vehicle classes is available here: http://driveubernyc.com/vehicles/full-list.

[5] Hall and Krueger (2016) compare Uber's driver-partners labor market demographic to taxi drivers. They find that Uber's driver-partners are more similar to the general workforce than taxi drivers and chauffeurs. In particular, Uber drivers are comparatively more educated and younger.

As there is no rain surcharge for taxis, taxi drivers may be less likely to respond to the increased demand for rides. The extra compensation that Lyft and Uber drivers receive through dynamic pricing may incentivize them to respond to increased earning opportunities with a greater magnitude than taxi drivers. Additional compensation may also incentivize Lyft and Uber drivers to overcome the disutility of working in adverse conditions. This is thus an ideal setting for studying intertemporal models of labor (Fehr and Goette, 2007).

2.3. Evaluating the effect of rain

We test five hypotheses in what follows. First, we test whether the number of Uber and Lyft rides increases significantly more than the number of taxi rides in rainy hours. Due to dynamic pricing, among other reasons, we expect that the number of rides given by Uber and Lyft drivers increases significantly more than the number of taxi rides in the rain. Such a result would indicate that Uber and Lyft drivers are more responsive to rain and that it is (at least partly) because of the increase in fare coming from dynamic pricing. The two types of drivers (Lyft and Uber, and taxi) could thus have similar labor supply elasticities, but react differently to rain due to different instant wage changes.

Second, we investigate whether rain increases the number of daily rides for taxis, Lyft and Uber. The neoclassical model predicts that rain increases significantly the number of daily rides. On the other hand, if drivers have a daily target income level, then we should find that rainy days are not significantly related to taxi, Lyft and Uber rides. Note that it is possible that drivers have a weekly or monthly target income level. Unfortunately, the sample size is too small to estimate precisely whether this is the case.

A key assumption for these two hypotheses is that adverse weather conditions identically increases demand for taxis, Lyft and Uber cars.[6] During rainfall, it is generally considered to be undesirable to be outside (Connolly, 2008). As such, individuals who would generally walk or wait for public transit may wish to utilize taxis, Lyft or Uber cars. A second assumption is that these periods of increased demand, proxied by rain, leads to surge pricing (prime time) on the Uber (Lyft) platforms. Anecdotal evidence suggests that this is the case.[7]

Note that explanations other than dynamic pricing could explain the larger effect of rain on Lyft and Uber rides. Customers who take Uber and Lyft may be different than individuals preferring taxis, e.g., different willingness-to-pay (WTP). Similarly, cab drivers and Uber and Lyft drivers may also be different. While driving is a primary occupation for most traditional cab drivers, it is a secondary job for many Uber drivers (Chen et al., 2017; Koustas, 2018). Arguably, hours supplied to a secondary job are more elastic than hours supplied to a primary job. Taxi and Uber drivers may also have a different aversion to driving in inclement weather.[8] We restrict the analysis to Manhattan to make customers and drivers more comparable. Moreover, we include a large set of time fixed effects (e.g., indicators for hour of day by day of week) in our model to make rides more comparable. Nonetheless, the response to weather conditions could be due to dynamic pricing, but also drivers' aversion to risk and customers' different WTP across platforms.

Taxi drivers also face a restricted supply of available taxi cabs. Lyft and Uber drivers are free to operate without acquiring a taxi medallion,[9] allowing for a more flexible, unconstrained supply of drivers. Lyft and Uber drivers may also freely choose when they work and they can drive as long as they like. Farber (2015) finds that the number of taxi cars operating decreased during rainy hours, indicating that the capacity constraints of taxi drivers to respond to the increased earning opportunity presented by rain is not binding.

The third hypothesis that we test is whether taxi drivers' supply response to rain changed since Uber entered the market. The lack of rain surcharge for taxis and the increasing presence of Uber drivers when it rains may increase the difficulty of finding customers for taxi drivers. We test this using taxi rides data pre- and post-Uber.

Fourth, we test whether the total (taxi, Lyft and Uber) number of rides increased or decreased since Uber entered the market and whether it was relatively easier to get a ride in rainy than in non-rainy hours in post-Uber years. Note that the presence of Uber does not automatically increase the total number of rides since there might be a substitution of rides from taxi to Uber. It thus remains unclear whether the total number of rides in the rain increased proportionally more than the total number of rides in non-rainy hours since Uber entered the market.

Finally, we test whether Lyft's entry in the market affected Uber. Uber and Lyft are direct competitors and are both subject to dynamic pricing.[10]

A key assumption for the third, fourth and fifth hypotheses is that the increase in demand for rides during rainy hours is similar in pre- and post-Uber (Lyft) years. We control for demand-side factors such as the natural log of population (18 years and over) in our empirical analysis.

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[6] The New York subway provided significantly more rides than Uber and taxi drivers. Subway and bus ridership statistics are available here: http://web.mta.info/nyct/facts/ridership/.

[7] The following website provides surge pricing data for Uber: http://uberestimate.com/prices/Manhattan/. We chose popular pickup starting points and looked at whether there was surge pricing when it rained during April 2018. We find that rain led to surge pricing most of the time.

[8] Hall and Krueger (2016) find that there is a greater representation of younger people among Uber drivers than among taxi drivers.

[9] Taxis are required to purchase one of the city's prized taxi medallions in order to operate. The supply of taxi medallions (and thus taxis) is limited. The cost of a taxi medallion is often hundreds of thousands of dollars.

[10] Anecdotal evidence suggests that some Uber drivers are also Lyft drivers.

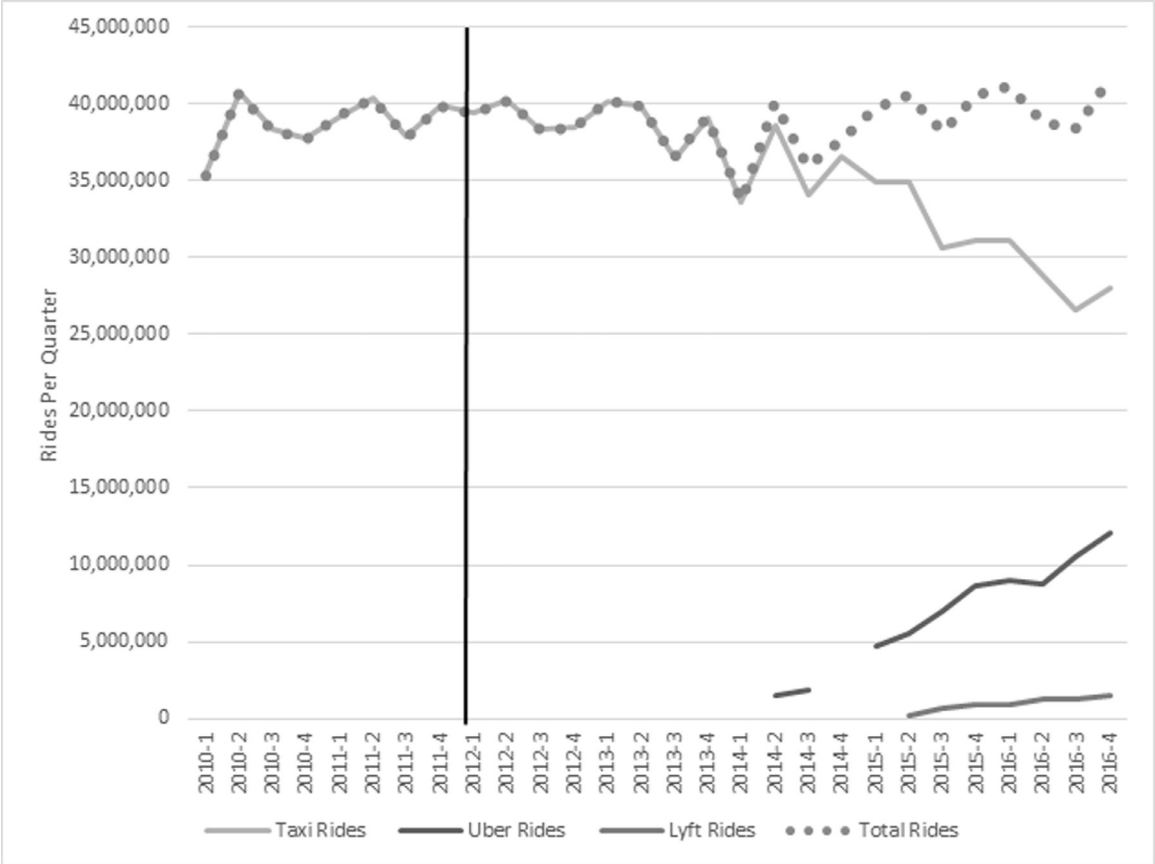

Fig. 1. Number of taxi, Lyft, Uber and total (taxi, Lyft and Uber) rides per quarter. Uber entered the market in May 2011. Lyft entered the market in July 2014.

3. Data sources

3.1. Data on taxi, Lyft and Uber rides

This study relies on all trip level data of Uber pickups in NYC from April to September 2014 and January 2015 to December 2016. Additionally, all trip level data of Lyft pickups from April 2015 to December 2016 is utilized. This data set was made available by the New York Taxi Commission (TLC). This data set includes approximately 110 million Uber rides and approximately 14 million Lyft rides, as reported to the TLC by the respective dispatching bases utilized by each provider in NYC. Both Lyft and Uber cars are classified as for-hire vehicles by the TLC.[11]

Individual trip level data for taxi rides are also available for this time frame. TLC has publicly released trip level data for all yellow taxi rides dating back to 2009. This is when the TLC started to require that digital trip records be kept through car GPS devices. For this time period, we calculated the number of rides that occurred hourly for Lyft, Uber and taxis. This was done using the time of the pick up. For taxis, we also compute the hourly number of passengers and the total time-and-distance fare as calculated by the taxi meter per hour. Note that taxi rides going to the airport are excluded for these two additional variables. The total time-and distance fare does not include extra charges, such as the rush hour ($1) and overnight charges ($0.50), the $0.50 MTA tax, or the $0.30 improvement surcharge that started in 2015. It also does not include tip amount, or tolls amount. Unfortunately, we cannot compute these additional variables for Lyft or Uber since we do not observe the price of each ride nor any fare multiplier (surge price or prime time.)

Fig. 1 shows the number of rides per quarter since 2009 for taxis, over the period April to September 2014 and January 2015 to December 2016 for Uber and over the period April 2015 to December 2016 for Lyft. Fig. 1 also plots the total (taxi, Lyft and Uber) number of rides. Note that Uber (Lyft) entered the New York market in May 2011 (July 2014). What emerges is that the number of taxi rides is much larger than the number of Uber rides in 2014. During the second quarter of 2014,

[11] TLC publishes Lyft and Uber data aggregated with other for-hire vehicles (FHVs). Only Lyft and Uber data is used for this report; observations from any other FHV company were removed.

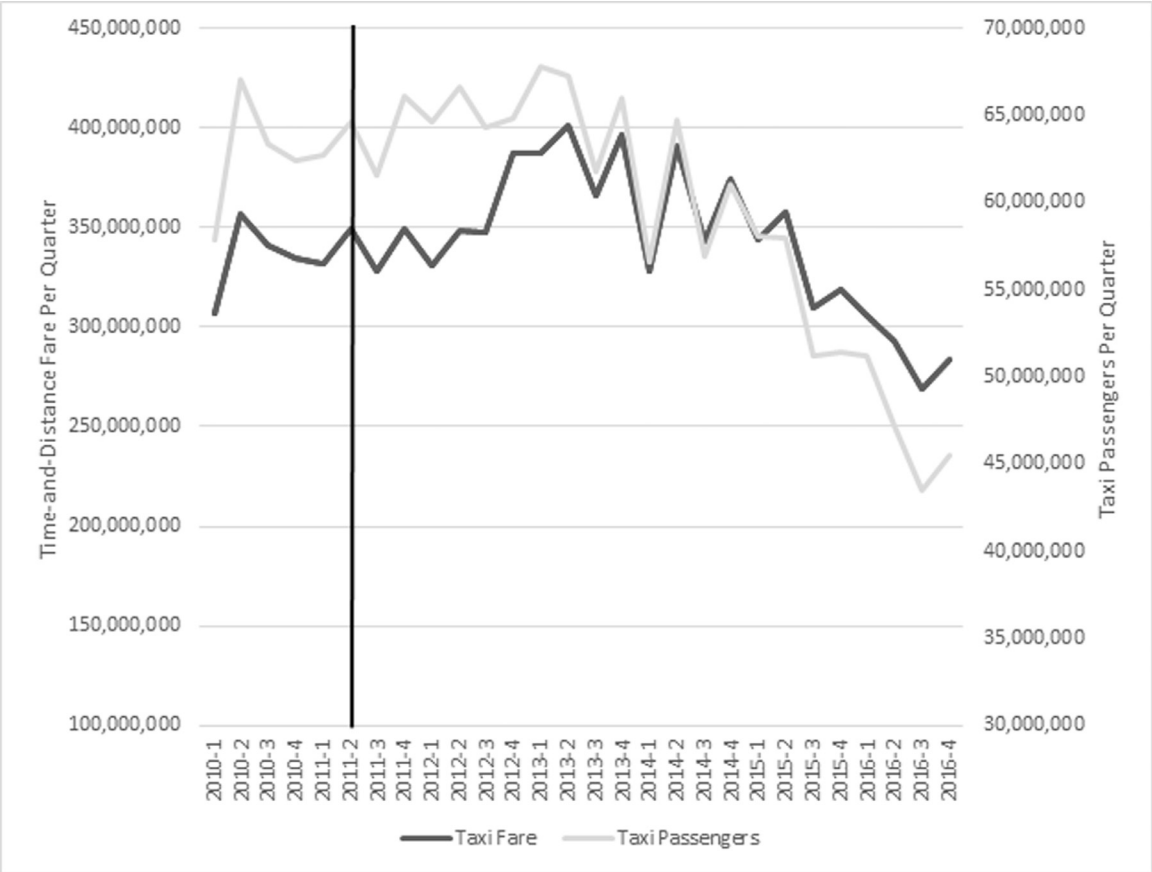

Fig. 2. Number of number of taxi passengers and total time-and-distance fare (in 2010 dollars) as calculated by the taxi meter per quarter. Uber entered the market in May 2011. Lyft entered the market in July 2014.

when Uber had been in the market for nearly three years, there were approximately 23 times more taxi rides than Uber rides. But this gap has decreased sharply. During the last quarter of 2016, the most recent quarter in our analysis, there were approximately 1.4 times more taxi rides than Uber rides. Fig. 1 suggests that this decreasing gap is due both to an increase in Uber rides and a decrease in taxi rides. During this same quarter, Lyft has a much smaller share of rides: there were approximately 9 times more taxi rides than Lyft rides, and approximately 6 times more Uber rides. Fig. 2 displays the number of taxi passengers and total time-and-distance fare as calculated by the taxi meter per quarter for the period 2010–2016. The number of passengers per quarter for taxis decreased by approximately 16 million passengers from 2010 to 2016 while total time-and-distance fare per quarter decreased by approximately $30 million in 2010 dollars.[12]

The differential in rides during rainy hours also decreased during this time period. During the second and third quarters of 2014, there were respectively 3.9 million taxi rides and 236,000 Uber rides during rainy hours (i.e., 16 times more taxi rides than Uber rides). During the second and third quarters of 2016, the gap was down to 1.2 million, with 1.4 million Uber rides and 2.6 million taxi rides (i.e., less than two times more taxi rides than Uber rides in rainy hours).

For our analysis, two subsets of the taxi data set are used. A first taxi data from April to September 2014 and January 2015 to December 2016 (2014–2016 henceforth) is used, as it is a direct comparison to the time frame for which Uber data are available. A second subset, from January 2010–April 2011 (2010–2011 henceforth), is used as well. We rely on this second subset which covers a period before Uber entered the NYC market as a comparison for taxi drivers' response in post-Uber market conditions.

We provide descriptive statistics for taxi, Lyft and Uber rides per hour in Panel A of Table 1. (See Appendix Table A1 for taxi, Lyft and Uber rides per day.) The average number of Uber rides per hour is 3246 over the period 2014–2016, while the average number of Lyft rides is 448. During the same time period, the average number of taxi rides per hour was 14,878. This is a sizable decrease in comparison to the average number of taxi rides over the period 2010–2011 (17,704). The standard deviation of taxi rides has also decreased in this same time frame. Panels B and C provide the same summary statistics as in Panel A, but for two sub-samples: 1) excluding nighttime observations (1:00–5:00 a.m.), and 2) excluding

[12] We use the Consumer Price Index for All Urban Consumers (All items) in New York-Northern New Jersey-Long Island to deflate prices.

Table 1
Summary statistics (per Hour).

	Mean	Std. Dev.	Max	Min	Obs.
Panel A					
Uber rides	3246	2551	16,496	5	21,423
Lyft rides	448	347	1998	3	14,886
Taxi rides (Post-Uber)	14,878	6904	32,435	183	21,423
Taxi rides (Pre-Uber)	17,704	8610	48,250	171	11,551
Panel B	Dropped night (1:00–5:00 a.m.)				
Uber rides	3780	2535	15,750	5	16,973
Lyft rides	511	348	1998	3	11,793
Taxi rides (Post-Uber)	17,331	5110	32,435	183	16,973
Taxi rides (Pre-Uber)	20,595	6714	48,250	343	9133
Panel C	Dropped winter (January–March)				
Uber rides	3258	2639	16,496	28	17,125
Lyft rides	449	359	1998	5	12,740
Taxi rides (Post-Uber)	14,751	6790	32,435	1163	17,125
Taxi rides (Pre-Uber)	17,887	8493	48,250	171	7264

Note: Authors' calculations. We report rides per hour. See Data section for more details. The time period pre-Uber is January 2010-April 2011. The period post-Uber is April–September 2014 and 2015–2016. The time period for Lyft is April 2015–2016. Panel B excludes from the sample rides from 1:00 to 5:00 a.m.. Panel C excludes from the sample the months of January, February and March.

winter months (January–March). The average number of rides per hour is higher when we exclude nighttime observations and the number of taxi, Lyft and Uber rides is somewhat similar during winter and non-winter months.[13]

3.2. Weather data

The above data sets are combined with weather data from the National Weather Service Observatory in Central Park. This data set was collected using scraping techniques from www.wunderground.com. The observatory reports observations at 51 min past the hour. As such, when the number of rides per hour were calculated, the number of taxi trips occurring in the half hour preceding and following the weather observation was aggregated into a single observation. For example, for a 10:51 a.m. weather observation, the number of rides occurring from 10:21:00 a.m. to 11:20:59 a.m. would be counted. Each observation reports the weather condition of the time period. Possible conditions include "Clear", "Heavy Rain", "Rain", "Light Rain" or "Partly Cloudy". A full list of weather conditions can be seen in Table 2.

Comparing the number of different weather condition observations from 2010 to 2011 to 2014–2016,[14] we find that there is a somewhat similar proportion of instances of each condition. The difference is not statistically significant at conventional levels for both "Heavy Rain" and "Light Rain." The condition is coded as "Heavy Rain" in 0.3% of the observation in the time period 2014–2016 and the time period 2010–2011. Similarly, there was "Light Rain" in 3.3% of the observation also in both time periods. The condition is coded as "Rain" slightly more often in the post-Uber period. Overall, the difference in the amount of rain ("Heavy Rain', "Rain", "Light Freezing Rain" or "Light Rain") for the periods pre- and post-Uber is not statistically significant at the 10% level.[15]

4. Identification strategy

In this section, we describe the main specification and the controls. The objective is to investigate the impact of an increase in demand, identified by rain, on taxi, Lyft and Uber rides. To identify this effect, we use trip level data to calculate the number of rides for each of the clock hours in the time period analyzed. Our unit of observation is an hour. We use only rides inside Manhattan.[16]

In our main specification, we estimate:

$$Y_{hmy} = \alpha + \beta RAIN_{hmy} + \delta_m + \gamma_y + X'_{hmy}\lambda + \varepsilon_{hmy}, \tag{1}$$

[13] Appendix Figs. 3–5 confirm that the fraction of rides is lower for nighttime observations by showing the fraction of rides per hour of day for taxis, Lyft and Uber (post-Uber years). The fraction of Lyft and Uber rides from 5:00 to 10:00 p.m. is slightly larger than the fraction of taxi rides while this is the opposite from 8:00 a.m. to 1:00 p.m.

[14] See Appendix Table A2 for a comparison of weather events for the time periods pre- and post-Lyft. Similarly, we find that there is a somewhat similar proportion of instances of each weather condition.

[15] There are also some cases where the observatory does not report the conditions at the 51st min. The number of observations where this is the case is 28 in 2014–2015. The observatory also has 51 cases of instances where the reported weather condition was "unknown" in 2014–2015. These observations are removed from the analysis.

[16] We think using only rides inside Manhattan reduces measurement error in the weather variable and makes customers for Uber and taxis more similar. Note that the National Weather Service Observatory is in Central Park.

Table 2
Weather data.

	Post-Uber period	Pre-Uber period	Difference
Two-sided *t*-tests			
Heavy rain	56 (0.3%)	31 (0.3%)	−0.001 (0.006)
Rain	155 (0.7%)	111 (0.9%)	−0.024 (0.010)
Light rain	711 (3.3%)	385 (3.3%)	−0.001 (0.021)
Light freezing rain	11 (0.1%)	14 (0.2%)	−0.007 (0.003)
Heavy snow	8 (0.0%)	18 (0.2%)	−0.012 (0.003)
Snow	53 (0.3%)	55 (0.5%)	−0.022 (0.007)
Light snow	120 (0.6%)	130 (1.1%)	−0.057 (0.010)
Clear	11,496 (53.7%)	6166 (53.4%)	0.028 (0.058)
Scattered clouds	901 (4.2%)	438 (3.8%)	0.041 (0.023)
Partly cloudy	1,244 (5.8%)	616 (5.3%)	0.047 (0.027)
Mostly cloudy	1,656 (7.7%)	805 (7.0%)	0.076 (0.030)
Overcast	4509 (21.1%)	2611 (22.6%)	−0.016 (0.047)
Fog	5 (0.0%)	4 (0.1%)	−0.001 (0.002)
Haze	495 (2.3%)	167 (1.5%)	0.086 (0.016)
Mist	2 (0.0%)	0 (0.0%)	0.001 (0.001)
Total	21,422 (100%)	11,551 (100%)	

Note: Data from the National Weather Service Observatory in Central Park. The time period pre-Uber is January 2010-April 2011. The period post-Uber is April–September 2014 and 2015–2016. Percentages are in parentheses (standard errors for the last column).

where Y_{hmy} is the log of the number of Uber, Lyft or taxi rides in hour h, month m and year y. $RAIN_{hmy}$ is a dummy that equals one if it rained in Central Park in hour h and zero otherwise. More precisely, $RAIN_{hmy}$ is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain". We include month and year fixed effects and X_{hmy} is a vector of other regressors that includes indicators for hour of day by day of week, snow and major holiday.[17] The coefficient of interest here is β.

We check the robustness of our results by removing outliers.[18] First, we check that the association of rain and the number of rides per hour is robust to using a subset of the data set that removes the winter months. The winter months are identified in this research as January, February and March. Second, we exclude nighttime observations since the number of rides is lower during that time of the day. Last, we test the robustness of our findings by excluding from the analysis the more severe weather events "Heavy Rain" and "Light Freezing Rain".

Our econometric model for testing the effect of Uber's rise in popularity on the number of taxi rides in rainy hours is as follows:

$$Y_{hmy} = \alpha + \beta RAIN_{hmy} + \theta PostUber_{hmy} + \sigma RAIN_{hmy} \times PostUber_{hmy} + \delta_m + X'_{hmy}\lambda + Z'_{my}\mu + \varepsilon_{hmy}, \quad (2)$$

where the dependent variable is the log of the number of taxi rides in an hour h, month m and year y. *PostUber* is a variable that takes the value 1 if the year is after 2011. The interaction of *RAIN* and *PostUber* shows the effect of the introduction of Uber's platform in NYC on taxi rides in rainy hours. We include month fixed effects, our vector of time dummies (X_{hmy}) and control for demand-side factors such as the natural log of population (18 years and over) at the year level and the monthly unemployment rate (Z_{my}).

The interpretation relies on the identification condition that there are no other time-varying shocks that affect the number of taxi rides.[19] We test the robustness of our results by interacting our month fixed effects (δ_m) with the indicator *PostUber*. We also include a linear time trend.

Last, we test whether Lyft's entry affected taxi and Uber rides. In this econometric model, we simply replace the variable *PostUber* by the variable *PostLyft*. *PostLyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise.

[17] The dummy for snow is equal to one if the conditions were noted as "Snow", "Heavy Snow" or "Light Snow" and zero otherwise. Major holidays include New Years Day, Martin Luther King Day, Lincoln's Birthday, President's Day, Memorial Day, Independence Day, Labor Day, Columbus Day, Veterans Day, Thanksgiving and Christmas Day.

[18] We check whether the inclusion of historic weather patterns affects our main conclusions. For instance, our results are robust to adding as a control a dummy for whether it rained the previous year, e.g., whether it rained on December 31st, 2014 for all the observations (hours) on December 31st, 2015. Results available upon request.

[19] Green taxi cabs (i.e., Boro taxis), which are able to serve the boroughs of New York and drop passengers off in the Manhattan core, began operating in 2013. These cabs are unable to pick up trips in the south of Manhattan. We thus exclude Green taxi cabs from the analysis. We replicate our analysis combining rides in Manhattan and outside Manhattan in the Appendix. We include Green taxis for this exercise.

Table 3
Effect of rain on taxi, Lyft and Uber rides per hour: post-Uber.

	Natural log oftTaxi Rides					
	(1)	(2)	(3)	(4)	Exclude night (5)	Exclude winter (6)
Panel A						
Rain	0.014	0.011	0.001	0.046	0.047	0.049
	(0.035)	(0.035)	(0.034)	(0.007)	(0.007)	(0.007)
Observations	21,423	21,423	21,423	21,423	16,973	17,125
R-squared	0.000	0.006	0.035	0.953	0.866	0.966
	Natural log of Uber rides					
	(1)	(2)	(3)	(4)	Exclude night (5)	Exclude winter (6)
Panel B						
Rain	0.179	0.178	0.194	0.224	0.253	0.234
	(0.068)	(0.068)	(0.043)	(0.017)	(0.018)	(0.020)
Observations	21,423	21,423	21,423	21,423	16,973	17,125
R-squared	0.001	0.002	0.497	0.934	0.921	0.942
	Natural log of Lyft rides					
	(1)	(2)	(3)	(4)	Exclude night (5)	Exclude winter (6)
Panel C						
Rain	0.148	0.148	0.101	0.192	0.218	0.189
	(0.085)	(0.085)	(0.053)	(0.033)	(0.037)	(0.038)
Observations	14,886	14,886	14,886	14,886	11,793	12,740
R-squared	0.001	0.001	0.463	0.852	0.817	0.849
Control variables						
Hour × day FE				✓	✓	✓
Week FE			✓	✓	✓	✓
Month & year FE			✓	✓	✓	✓
Holiday & snow FE		✓	✓	✓	✓	✓

Note: The unit of observation is an hour. The dependent variables are respectively the natural log of the number of taxi rides (Panel A), the natural log of the number of Uber rides (Panel B) and the natural log of the number of Lyft rides (Panel C). The time period is April–September 2014 and 2015–2016 in Panels A and B and April 2015–2016 in Panel C. *Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" and zero otherwise. Column 5 excludes from the sample rides from 1:00 to 5:00 a.m. Column 6 excludes from the sample the months of January, February and March. Standard errors clustered by unique day are in parentheses.

5. Results

5.1. Impact of rain on hourly rides post-Uber

We first estimate the effect of rain on taxi rides per hour. We then test whether the effect of rain is similar for the number of Uber and Lyft rides. If the demand is unmet in rainy hours, then Uber and Lyft dynamic pricing, combined with a more flexible supply of vehicles, could encourage an increase in supply.

Table 3 contains OLS estimates of Eq. (1) for Uber, Lyft and taxi rides. The time period is April–September 2014 and January 2015 to December 2016 periods (i.e., post-Uber period) for Uber and taxis and April 2015–2016 for Lyft. The dependent variables are respectively the natural log of the number of taxi rides (Panel A), the natural log of the number of Uber rides (Panel B) and the natural log of the number of Lyft rides (Panel C). The sample size is 21,423 observations (i.e., hours) in Panels A and B and 14,886 observations in Panel C. We report standard errors clustered by unique day (e.g., May 25, 2015, March 18, 2014, etc.) in parentheses.

What clearly emerges is that rainy hours are associated with a small increase in taxi rides and a large increase in Uber and Lyft rides. In column 1, we do not include any fixed effects and find that Uber rides increase by approximately 18% while taxi rides increase by only 1%. Lyft rides increase by slightly less than Uber rides. We include dummies for snow and major holidays in the remaining columns and our set of time fixed effects in columns 3–6. The number of Uber rides is about 22% higher when it is raining and the estimate is significant at the 1% level. (The number of Lyft rides increases by approximately 19%.) On the other hand, the number of taxi rides per hour increases by only 5%.

In column 5, we test the robustness of our estimates by excluding nighttime observations. The estimates are slightly larger for Uber and Lyft rides, but the difference is not statistically significantly different from the estimates including night-

Table 4
Robustness checks for the period post-Uber (per hour).

Natural log of	Taxi rides (1)	Taxi rides (2)	Uber rides (3)	Uber rides (4)	Lyft rides (5)	Lyft rides (6)
Light freezing rain	−0.071		0.158		−0.150	
	(0.006)		(0.043)		(0.212)	
Light rain	0.039		0.189		0.167	
	(0.006)		(0.015)		(0.034)	
Rain	0.078		0.331		0.271	
	(0.016)		(0.038)		(0.055)	
Heavy rain	0.071		0.380		0.339	
	(0.027)		(0.060)		(0.070)	
Precipitation (in mm)		0.032		1.018		0.803
		(0.088)		(0.132)		(0.189)
Control variables						
Hour × day FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Month & year FE	✓	✓	✓	✓	✓	✓
Holiday & snow FE	✓	✓	✓	✓	✓	✓
Observations	21,423	21,423	21,423	21,423	14,886	14,886
R-Squared	0.953	0.953	0.935	0.934	0.853	0.852

Note: The unit of observation is an hour. The dependent variables are respectively the natural log of the number of taxi rides (col. 1–2), the natural log of the number of Uber rides (col. 3–4) and the natural log of the number of Lyft rides (col. 5–6). *Light Freezing Rain* is a dummy that is equal to one if the condition was noted as "Light Freezing Rain" and zero otherwise. *Light Rain* is a dummy that is equal to one if the condition was noted as "Light Rain" and zero otherwise. *Rain* is a dummy that is equal to one if the conditions were noted as "Rain" and zero otherwise. *Heavy Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain" and zero otherwise. *Precipitation* is the amount of rain in millimeter. Standard errors clustered by unique day are in parentheses.

time observations. In column 6, we exclude the months of January, February and March. This specification check confirms that Uber and Lyft rides increase by approximately 20% in rainy hours while taxi rides increase by only 5%.[20]

In Table 4, columns 1, 3 and 5, we separately test the effect of light rain, light freezing rain, rain and heavy rain on rides per hour. The estimates are positive and statistically significant for the four variables when the dependent variable is the log of the number of Uber rides. The estimated coefficients are larger for *Rain* (coeff. 0.331, std. error 0.038) and *Heavy Rain* (coeff. 0.380, std. error 0.060) than for *Light Rain* (coeff. 0.189, std. error 0.015) suggesting that more rain leads to more Uber rides. The estimates are also positive and significant for Lyft rides for light rain, rain and heavy rain. The estimates are much smaller for taxi rides than for Uber and Lyft rides. The estimate is negative and significant for light freezing rain, but positive and significant for light rain, rain and heavy rain. The estimates range from 4% for light rain to 7% for rain and heavy rain.

In columns 2, 4 and 5, we replace our dummy rain variables by the variable *Precipitation*. The variable *Precipitation* captures rain intensity in a given hour. This variable is equal to the amount of rain in millimeter. We find a positive association between this variable and the number of Uber, Lyft and taxi rides, but the size of the estimates are much larger for Lyft and Uber than for taxi rides.

We replicate our analysis combining rides inside and outside Manhattan in Appendix Table A4. The structure of the table is the same as Table 3. Our estimates and conclusions for Uber, Lyft and taxis are very similar.[21]

So far, our results provide suggestive evidence that dynamic pricing encourages an overall increase in supply for Uber and Lyft drivers. In other words, rain may increase the number of rides for Uber and Lyft through their flexible pricing schemes. Rain has a different impact on taxis who are facing different instant wage changes than Lyft and Uber drivers, i.e., taxis are subject to a fixed price. Due to data constraints (i.e., unable to identify individual drivers or cars) we are unable to estimate an elasticity for Uber or taxi drivers. What is presented is an approximate semi-elasticity: the increase in total trips due to an increase in demand. Unfortunately, it is also impossible to test whether the increase in Uber and Lyft rides during rainy hours is due to an increase in hours worked (intensive margin) or an increase in the number of drivers working (extensive margin).

5.2. Impact of rain on daily rides post-Uber

In Table 5, we turn to presenting OLS estimates of the relationship between rain and *daily* rides. (See Appendix Table A6 for the other taxi outcomes at the day level.) The time period is the same as in Table 3 and the dependent variables

[20] Appendix Table A3 provides additional robustness checks. We exclude respectively hours of heavy rain and light freezing rain. The estimates are very similar to the estimates presented in Table 3 confirming that Uber and Lyft rides increase by significantly more than taxi rides in rainy hours.

[21] We also look at the effect of rain on Green taxi rides. Note that Green taxis are not allowed to pick up street hails in Manhattan below 110th St. on the West Side, and below 96th St. on the East Side. Rain increases the number of Green taxi rides by about 5–10%.

Table 5
Effect of rain on taxi, Lyft and Uber rides per day: post-Uber.

	Natural log of taxi rides					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A						
Rain day	0.0245	0.0250	0.0271			
	(0.0152)	(0.0148)	(0.0134)			
Hours of rain				0.0051	0.0047	0.0036
				(0.0022)	(0.0023)	(0.0019)
Observations	789	789	789	789	789	789
R-squared	0.002	0.033	0.337	0.003	0.033	0.336
	Natural log of Uber rides					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B						
Rain day	0.0373	0.0397	0.0994			
	(0.0714)	(0.0701)	(0.0322)			
Hours of rain				0.0187	0.0186	0.0237
				(0.0108)	(0.0106)	(0.0038)
Observations	789	789	789	789	789	789
R-squared	0.000	0.005	0.844	0.003	0.007	0.846
	Natural log of Lyft rides					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel C						
Rain day	−0.0059	−0.0072	0.0557			
	(0.1028)	(0.1029)	(0.0681)			
Hours of rain				0.0125	0.0127	0.0087
				(0.0171)	(0.0171)	(0.0102)
Observations	515	515	515	515	515	515
R-squared	0.000	0.002	0.699	0.001	0.003	0.699
Control variables						
Week FE			✓			✓
Month & year FE			✓			✓
Holiday & snow FE		✓	✓		✓	✓

Note: The unit of observation is a day. The dependent variables are respectively the natural log of the number of taxi rides (Panel A), the natural log of the number of Uber rides (Panel B) and the natural log of the number of Lyft rides (Panel C). The time period is April–September 2014 and 2015–2016 in Panels A and B and April 2015–2016 in Panel C. *Rain Day* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" at any point during the day and zero otherwise. *Hours of Rain* is a variable that is equal to the number of hours in which the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain". Robust standard errors are in parentheses.

are respectively the natural log of the number of taxi rides (Panel A), the natural log of the number of Uber rides (Panel B) and the natural log of the number of Lyft rides (Panel C). Our variable of interest in columns 1–3, *Rain Day*, is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" at any point during the day and zero otherwise. In columns 4–6, we replace *Rain Day* with the variable *Hours of Rain. Hours of Rain* is a continuous variable that is equal to the number of hours of rain per day. Note that the average number of hours of rain in a rainy day is 3.94 (std. deviation 3.40). We include holiday and snow dummies in columns 2, 3, 5 and 6 and week, month and year fixed effects in columns 3 and 6.

In our favorite specification (column 3), the estimates suggest that the number of Uber rides increases by approximately 10% during rainy days. The estimate is significant at the 1% level. We also find that an additional hour of rain increases the number of Uber rides per day by 2%. The result that rain increases daily rides provides evidence that Uber drivers do not have reference-dependent preferences in which workers have a reference level of daily income (Chen and Sheldon, 2015). There is also weak evidence that rain increases the number of taxi rides per day. The estimates are small, positive and statistically significant. Both Uber and taxi drivers thus demonstrate a response suggestive of a neoclassical intertemporal model of labor: the total number of rides per day increases with increased demand. We also find positive estimates for Lyft, but the sample size is smaller and the estimates are not statistically significant.

Table 6
Effect of rain on taxi rides per hour: pre- and post-Uber.

Natural log of	Taxi rides (1)	Taxi rides (2)	Taxi rides (3)	Taxi rides (4)	Taxi rides (5)	Taxi rides (6)
Post-Uber	−0.160	−0.259	−0.258	−0.259	−0.180	−0.464
	(0.009)	(0.094)	(0.094)	(0.094)	(0.106)	(0.125)
Rain			0.049	0.041	0.051	0.046
			(0.007)	(0.013)	(0.014)	(0.010)
Rain × post-Uber				0.012	0.003	
				(0.017)	(0.017)	
Post-Lyft						−0.082
						(0.039)
Rain × post-Lyft						0.013
						(0.015)
Control variables						
Hour × day FE	✓	✓	✓	✓	✓	✓
Week & month FE	✓	✓	✓	✓	✓	✓
Month × post-Uber					✓	
Month × post-Lyft						✓
Time trend					✓	✓
Holiday & snow FE	✓	✓	✓	✓	✓	✓
Population & UE rate		✓	✓	✓	✓	✓
Observations	32,974	32,974	32,974	32,974	32,974	32,974
R-Squared	0.923	0.929	0.929	0.929	0.933	0.934

Note: The unit of observation is an hour. The dependent variable is the natural log of the number of taxi rides. The time period is January 2010–April 2011, April–September 2014 and 2015–2016. *Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" and zero otherwise. *Post Uber* is a dummy that is equal to one for the period April–September 2014 and 2015–2016 and zero otherwise. *Post Lyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise. We control for the natural log of population (18 years and over) at the year level and the monthly unemployment rate in columns 2–6. We include a linear time trend (i.e., monthly time trend) to our model in columns 5 and 6. Standard errors clustered by unique day are in parentheses.

5.3. Impact of Uber on taxi rides and income

In this subsection, we investigate whether the number of taxi rides, passengers and hourly fare income decreased after Uber entered the market. We also test whether the estimated effects of rain on these outcomes are similar in the pre-Uber era.

In Table 6, we test explicitly whether the number of taxi rides per hour reacts more to rain in the pre-Uber period than in 2014–2016. (See Appendix Table A7 for the analysis at the day level.) Columns 1 and 2 first show the relationship between the number of taxi rides and the variable *PostUber*. Column 1 includes our time fixed effects. Column 2 adds the natural log of population (18 years and over) at the year level and the monthly unemployment rate in the model. Columns 1 and 2 show that the number of taxi rides per hour decreased by approximately 15–25% in the post-Uber period. This result suggests that the market entry of Uber led to a significant decrease in taxi rides.

Column 3 adds our dummy variable for rain, but without the interaction with *PostUber*. Over the full period of study, the number of taxi rides in rainy hours increases by roughly 5%. Column 4 presents estimates of Eq. (2) where we include *PostUber*, the dummy for rain and the interaction term.[22] The interaction term, σ, is positive, but not statistically significant at the 10% level. This means that the number of taxi rides in rainy hours has not decreased by more than in non-rainy hours since Uber entered the market. In column 5, we check the robustness of this result by interacting our month fixed effects (δ_m) with the indicator *PostUber*. We also include a linear time trend. The interaction term is small and statistically insignificant.

We now test whether the entry of Lyft in the market affected the number of taxi rides. Column 6 replaces the variable *PostUber* by the variable *PostLyft*. *PostLyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise.[23] The interaction term of the variables *PostLyft* and rain is small and statistically insignificant, confirming that taxis are not responding differently since the entrance of market competitors to rain.

Our results suggest that taxis do not compete more (or less) vigorously with competitors during times of additional demand. This finding is consistent with the idea that taxis face market frictions that disincentive responses to capitalize on the increased demand during rain.

[22] The interaction of *RAIN* and *PostUber* shows the additional effect of Uber entering the market on the variation of taxi rides in rainy hours. A negative estimate could suggest that there is a substitution of taxi rides to Uber rides, and especially so when it rains. A negative effect could also be indicative that that the disutility of driving in the rain for taxi drivers has increased since May 2011. If the increasing number of Uber drivers in rainy hours results in more competition to find passengers, this may decrease the time spent with a passenger and decrease the willingness to work in the rain for taxi drivers.

[23] While Lyft entered the NYC market in April 2014, data on the number of rides is only available starting April 2015. During April 2015 the share of Lyft rides is low with about 110 rides per hour. Thus, we expect Lyft did not have an influential market share during April 2014–April 2015.

Table 7
Effect of rain on taxi passengers and fare per hour: pre- and post-Uber.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Natural log of taxi passengers					
Post-Uber	−0.144	−0.163	−0.162	−0.163	−0.107	−0.376
	(0.010)	(0.112)	(0.112)	(0.119)	(0.125)	(0.156)
Rain			0.054	0.047	0.060	0.053
			(0.008)	(0.016)	(0.017)	(0.012)
Rain × post-Uber				0.011	−0.002	
				(0.020)	(0.020)	
Post-Lyft						−0.079
						(0.040)
Rain × post-Lyft						0.012
						(0.018)
R-squared	0.918	0.923	0.923	0.923	0.928	0.929
Observations	32,974					
Panel B	Natural log of time-and-distance fare					
Post-Uber	−0.025	−0.060	−0.059	−0.061	−0.000	−0.217
	(0.010)	(0.095)	(0.095)	(0.095)	(0.106)	(0.126)
Rain			0.044	0.029	0.040	0.034
			(0.007)	(0.013)	(0.013)	(0.010)
Rain × post-Uber				0.024	0.013	
				(0.016)	(0.016)	
Post-Lyft						−0.072
						(0.041)
Rain × post-Lyft						0.031
						(0.014)
R-squared	0.911	0.917	0.918	0.918	0.923	0.923
Observations	32,974					
Control variables						
Hour × day FE	✓	✓	✓	✓	✓	✓
Week & month FE	✓	✓	✓	✓	✓	✓
Month × post-Uber					✓	
Month × post-Lyft						✓
Time trend					✓	✓
Holiday & snow FE	✓	✓	✓	✓	✓	✓
Population & UE rate		✓	✓	✓	✓	✓

Note: The unit of observation is an hour. The dependent variable in Panel A is the natural log of the number of taxi passengers. The dependent variable in Panel B is the natural log of the total time-and-distance fare as calculated by the taxi meter per hour (in 2010 dollars). The time period is January 2010–April 2011, April–September 2014 and 2015–2016. *Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" and zero otherwise. *Post Uber* is a dummy that is equal to one for the period April–September 2014 and 2015–2016 and zero otherwise. *Post Lyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise. We control for the natural log of population (18 years and over) at the year level and the monthly unemployment rate in columns 2–6. We include a linear time trend (i.e., monthly time trend) to our model in columns 5 and 6. Standard errors clustered by unique day are in parentheses.

Table 7 shows the effect of Uber on the other two taxi outcomes. The dependent variables are respectively the natural log of the number of taxi passengers (Panel A) and the natural log of the total time-and-distance fare (in 2010 dollars) as calculated by the taxi meter per hour (Panel B). Column 1 presents the relationship between these outcomes and the variable *PostUber*. The association between the number of taxi passengers and the variable *PostUber* is significant at the 1% level and suggest that taxi passengers decreased by about 15% since May 2011. The estimates suggest that total time-and-distance fare decreased by approximately 3% in the post-Uber period.

Of note, though, our research focuses on a time period in which Uber was still growing. The analysis pre- and post-Uber could yield very different conclusions when Uber reaches its steady state. For instance, the negative impact on hourly fare income is much bigger if we exclude the year 2014 (coeff. −0.186, std. error 0.010). This is consistent with Fig. 2 and with the idea that Uber was growing exponentially during the period 2014–2016.

We also present the estimated effects for the interaction term σ in Table 7, column 3. Our estimates are statistically insignificant for the two outcomes in column 4 suggesting that the number of passengers and total time-and-distance fare in rainy hours have not decreased by more than in non-rainy hours since Uber entered the market.

Table 8
Effect of Rain on Total (Taxi, Lyft and Uber) Rides per Hour: Pre- and Post-Uber.

Natural log of	Total Rides (1)	Total Rides (2)	Total Rides (3)	Total Rides (4)	Total Rides (5)	Total Rides (6)
Post-Uber	0.060	−0.084	−0.082	−0.085	−0.219	−0.238
	(0.009)	(0.089)	(0.089)	(0.089)	(0.102)	(0.124)
Rain			0.071	0.047	0.053	0.051
			(0.007)	(0.014)	(0.015)	(0.010)
Rain × Post-Uber				0.039	0.031	
				(0.017)	(0.017)	
Post-Lyft						−0.057
						(0.037)
Rain × Post-Lyft						0.097
						(0.010)
Control Variables						
Hour × Day FE	✓	✓	✓	✓	✓	✓
Week & Month FE	✓	✓	✓	✓	✓	✓
Month × Post-Uber					✓	
Month × Post-Lyft						✓
Time Trend					✓	✓
Holiday & Snow FE	✓	✓	✓	✓	✓	✓
Population & UE Rate		✓	✓	✓	✓	✓
Observations	32,974	32,974	32,974	32,974	32,974	32,974
R-Squared	0.929	0.930	0.930	0.930	0.932	0.932

Note: The unit of observation is an hour. The dependent variable is the natural log of the number of total (taxi, Uber and Lyft) rides. The time period is January 2010–April 2011, April–September 2014 and 2015–2016. *Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" and zero otherwise. *Post Uber* is a dummy that is equal to one for the period April–September 2014 and 2015–2016 and zero otherwise. *Post Lyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise. We control for the natural log of population (18 years and over) at the year level and the monthly unemployment rate in columns 2–6. We include a linear time trend (i.e., monthly time trend) to our model in columns 5 and 6. Standard errors clustered by unique day are in parentheses.

5.4. Impact of uber on total (Lyft, Uber and taxi) Rides

Turning now to a different dependent variable, Table 8 shows the effect of Uber entering NYC's market on the total number of rides (i.e., taxi, Lyft and Uber rides) per hour. (See Appendix Table A8 for the analysis at the day level.) The layout is the same as in Table 6. The coefficients on the *PostUber* variable indicate that the total number of rides per hour (and per day) remained unchanged in the time period 2014–2016. As shown before, this result is mostly due to the large increase in Uber rides and the large decrease in taxi rides.

The coefficient estimates for the rain dummy indicate that over the full period of study total rides increases by approximately 7.1% when it rains. Then, the next row shows the coefficient estimates for the interaction of the variables *RAIN* and *PostUber*. The estimated coefficient is positive and statistically significant at the 1% level, suggesting that, since Uber entered the market, it is relatively easier to get a ride in rainy than in non-rainy hours. More precisely, we find that the number of total rides in rainy hours increased by 3.9% in post-Uber years.[24] In column 5, we check whether this result is robust by interacting our month dummies and the indicator *PostUber*. We also include a linear time trend. The coefficient of interest remains positive and significant, suggesting that the result presented in column 4 is robust.

In column 6, we test whether the entry of Lyft in the market affected the total number of rides in rainy hours. Our estimate for the interaction term is positive and statistically significant at the 1% level. We find that the number of total rides in rainy hours increased by 10% in post-Lyft years. This result confirms that the entry of Lyft and Uber in the market made it easier to get a ride in the rain.

To sum up, the results are consistent with a substitution from taxi rides to Uber and Lyft rides and suggest that dynamic pricing combined with a more flexible supply of vehicles encourages an increase in supply for Uber and Lyft drivers.

5.5. Impact of Lyft on Uber rides

We conclude our analysis by testing whether the entry of Lyft in the market affected Uber drivers. In Table 9, we test whether the number of Uber rides in rainy hours decreased since Lyft entered the market. (See Appendix Table A10 for the analysis at the day level.) The time period is April–September 2014 and 2015–2016. The dependent variable is the natural log of the number of Uber rides. The layout is the same as in Table 6.

[24] In Appendix Table A9, we test whether the effect is significant for interaction of the variables *Light Rain* and *PostUber* and *Rain* & *Heavy Rain* and *PostUber*. The estimates are both positive and statistically significant.

Table 9
Effect of rain on Uber rides per hour: pre- and post-Lyft.

Natural log of	Uber rides (1)	Uber rides (2)	Uber rides (3)	Uber rides (4)	Uber rides (5)
Post-Lyft	1.355	0.357	0.360	0.296	0.288
	(0.024)	(0.024)	(0.023)	(0.089)	(0.102)
Rain			0.233	0.363	0.287
			(0.019)	(0.023)	(0.043)
Rain × post-Lyft				−0.091	−0.084
				(0.049)	(0.045)
Control variables					
Hour × day FE	✓	✓	✓	✓	✓
Week & month FE	✓	✓	✓	✓	✓
Month × post-Lyft					✓
Time trend					✓
Holiday & snow FE	✓	✓	✓	✓	✓
Population & UE rate		✓	✓	✓	✓
Observations	21,423	21,423	21,423	21,423	21,423
R-Squared	0.875	0.920	0.922	0.922	0.938

Note: The unit of observation is an hour. The dependent variable is the natural log of the number of Uber rides. The time period is April–September 2014 and 2015–2016. *Rain* is a dummy that is equal to one if the conditions were noted as "Heavy Rain", "Rain", "Light Freezing Rain" or "Light Rain" and zero otherwise. *Post Lyft* is a dummy that is equal to one for the period April 2015–2016 and zero otherwise. We control for the natural log of population (18 years and over) at the year level and the monthly unemployment rate in columns 2–5. We include a linear time trend (i.e., monthly time trend) to our model in column 5. Standard errors clustered by unique day are in parentheses.

Our estimates suggest that Uber was still growing after Lyft entered the market. Since April 2014, hourly Uber rides increased by about 35%. This result is consistent with Fig. 1. On the other hand, we find that Uber rides during rainy hours decreased by about 9% after the entry of Lyft. The estimates are robust to interacting our month dummies and the indicator *PostLyft* and to the inclusion of a linear time trend.

These results are consistent with the idea that Lyft and Uber are direct competitors in Manhattan. Lyft and Uber are growing exponentially during the years 2015–2016 and are decreasing the number of taxi rides. Lyft and Uber drivers are both subject to dynamic pricing and thus compete for rides when demand suddenly increases, i.e., during rainy hours.

6. Conclusion

Using all Uber and Lyft rides in NYC, we investigated the effect of rain on Uber and Lyft rides and found that the number of Uber (Lyft) rides per hour increases by approximately 22 (19)% when it is raining. During the same time period, the number of taxi rides per hour increases significantly less than the number of Uber and Lyft rides in rainy hours. We also provided evidence that during rainy days, the number of rides increases by about 10% for Uber. On the other hand, the number of taxi rides per day is only weakly related to rain. That rain would have a different effect on taxi and Lyft and Uber rides is in line with a neoclassical intertemporal model of labor, given that only Uber and Lyft rides are subject to dynamic rain pricing.

Looking at the impact of Uber on taxis, we found that the number of taxi rides per hour decreased by 26% after Uber entered the New York market in May 2011. There was also a significant decrease in hourly fare income and the number of passengers per hour. Notably, the response of taxis to rain has not changed since Uber entered the NYC personal transportation market. These results are consistent with the market frictions that taxi drivers face: the absence of dynamic pricing discourages a substantial response to increased demand.

We then provided evidence that Lyft and Uber are competing for rides during rainy hours. Since Lyft entered the market, Uber rides during rainy hours decreased by about 9%. This result is consistent with the idea that rain affects similarly Uber and Lyft drivers who are both subject to dynamic pricing during times of high demand.

Our results have important implications for the ongoing debate on whether Uber and Lyft are depressing taxi demand and whether Uber and Lyft increase consumers' welfare (Cohen et al., 2016). In particular, they highlight that Uber and Lyft are substituting for taxi cabs in NYC and that dynamic pricing seems effective in increasing labor supply during times of high demand.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.jebo.2018.06.004 .

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